

**The Nexus of Macroeconomic Policy Quality on Inflation: An Examination on  
Government Regulation and Its Effect on Consumers**

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**Abstract**

This paper investigates the relationship between the quality of a nation's macroeconomic policy, as measured by the World Bank's CPIA Macroeconomic Management Index, and its inflation rate. Using a panel dataset of 75 countries from 2005 to 2016, we estimate several fixed effects models to examine this relationship. Our results indicate that while higher macroeconomic policy ratings are associated with lower inflation rates, this relationship is dependent on the country's export volume. Furthermore, macroeconomic policy quality only achieves statistical significance after accounting for domestic financial sector credit growth and broader structural factors, including social equity, debt management, and property rights.

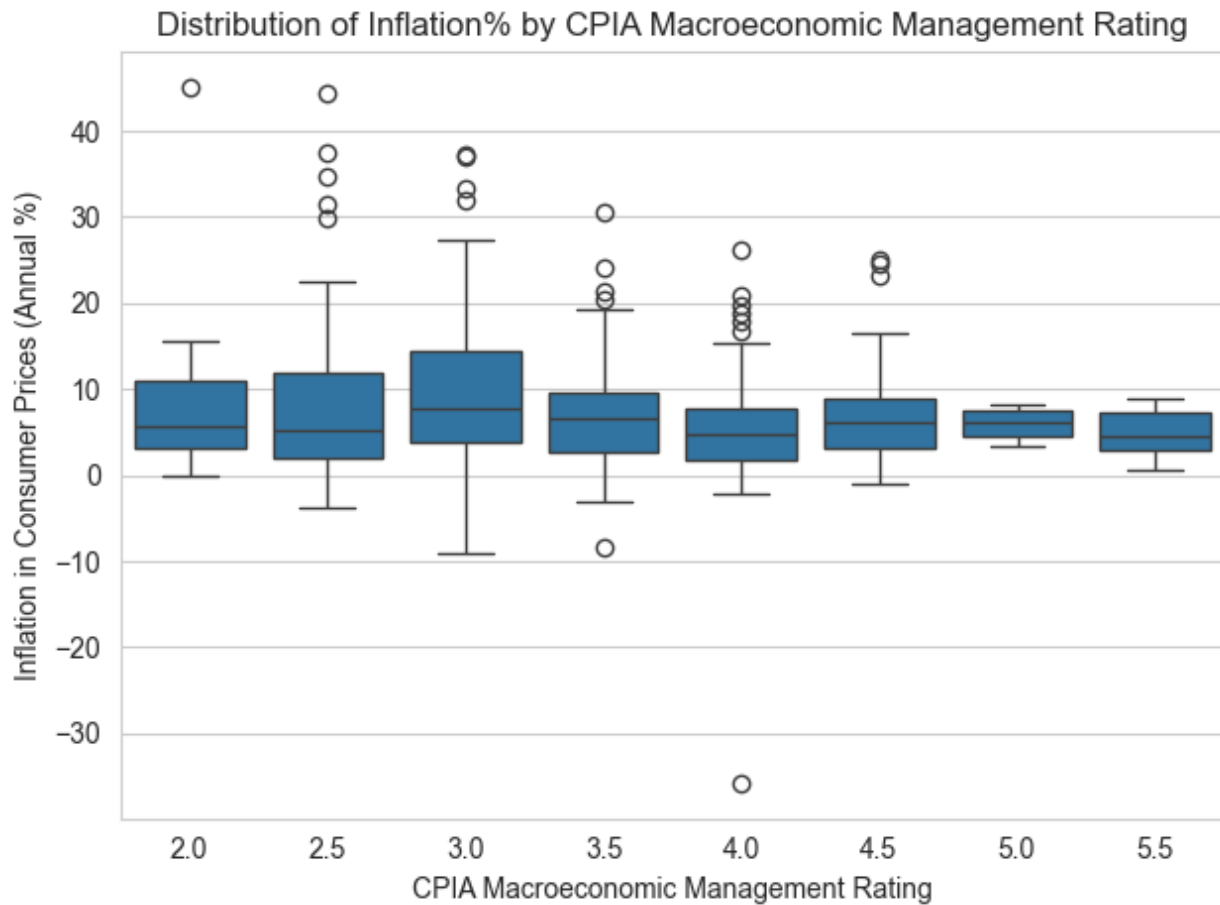
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### **Introduction**

In the lead-up to the 2024 U.S. presidential election, inflation emerged as a paramount concern among voters. A Pew Research Center survey conducted in May 2024 found that 62% of Americans considered inflation a “very big” problem for the country (Nadeem, 2024), and many Americans considered an administration’s proposed economic policy as a key factor in their voting decisions. This study investigates the role of macroeconomic management quality in shaping inflation outcomes, focusing on whether stronger macroeconomic policy, as measured by the World Bank’s CPIA Macroeconomic Management Index, reliably suppresses inflation. In contrast to prior work that often links inflation outcomes to broader governance or monetary policy stances, we explicitly test whether the effectiveness of macroeconomic quality is contingent on external factors such as trade exposure.

Though stronger macroeconomic management is traditionally expected to stabilize inflation, the substantial spread in inflation outcomes across varying policy quality levels, visualized in Figure 1, indicates that additional factors likely shape inflation dynamics, motivating the specific regression techniques used in this report. This paper finds that the impact of a nation’s macroeconomic management quality is associated with an expected decrease in inflation, but is dependent on that nation’s export volumes, controlling for broader institutional and financial sector quality.

Figure 1: Variance of Inflation % with respect to CPIA Macroeconomic Management



### Literature Review

Existing research is lacking in its investigation of the isolated effect of macroeconomic management on consumer price inflation. However, economic theory and existing empirical research identify several pathways through which macroeconomic quality shapes inflation.

Taken from classical economics, the Phillips curve demonstrates the relationship between economic output and inflation, with faster GDP growth or lower unemployment leading to upward price pressures (Phillips, 1958; Friedman, 1968). Modern empirical

findings confirm that demand-side pressures intensify inflation, potentially necessitating GDP controls. Income levels, proxied by GDP per capita, also modulate inflation dynamics. Developed economies exhibit stable, low inflation rates, whereas developing nations often experience greater volatility (Ha, Kose, & Ohnsorge, 2019). Credit growth similarly plays a vital role; rapid financial expansion can fuel demand and thus inflation (Choi & Cook, 2002). Prior research has also found that export booms or trade price volatility affect domestic inflation via terms-of-trade mechanisms (Frankel & Orszag, 2001). Strong banking systems and regulatory frameworks help stabilize inflation (Mishkin, 2007), whereas corruption and weak institutions exacerbate volatility (Salahodjaev & Chepel, 2014). Finally, infrastructure deficiencies create supply bottlenecks and can further destabilize prices (Straub, 2011).

Together, these factors demonstrate that inflation outcomes are shaped by a complex interplay of economic development, financial conditions, external shocks, and institutional quality. Accordingly, any empirical analysis seeking to isolate the effect of macroeconomic management on inflation must carefully control for these confounding influences. Therefore, controlling for GDP growth, income levels, credit conditions, trade shocks, financial sector depth, governance quality, and infrastructure is critical to isolating the impact of macroeconomic management on inflation. This study investigates each of these as potential sources of bias by including statistically significant variables in the finalized regression model.

### **Data Description**

*Table 1: Variable Descriptions*

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inflation       | Inflation in consumer prices (annual %)                                                                                                               |
| <i>Macro</i>    | CPIA macroeconomic management rating (1=low to 6=high)                                                                                                |
| <i>CPIA Avg</i> | A composite index calculated as the mean of CPIA ratings for economic management, social inclusion, financial sector development, and property rights |
| <i>Export</i>   | Export Volume Index (Standardized in comparison to 2,000)                                                                                             |
| <i>Credit</i>   | Domestic Credit provided to financial sector (% of GDP)                                                                                               |

This research employs panel data collected between 2005 and 2016 to examine the relationship between macroeconomic policy quality and inflation rates. The dataset contains 922 observations, covering 75 countries across a diverse range of economic, institutional, and developmental contexts. One observation is taken per country per year.

The primary dataset is the World Bank's Country Policy and Institutional Assessment (CPIA) database (World Bank, 2025). The CPIA provides ratings across several dimensions of governance and economic management. This paper focuses on the quality of macroeconomic policy formulation and implementation. The key independent variable, the CPIA Macroeconomic Management Rating, is drawn directly from this database and evaluated on a standardized scale ranging from 1 to 6.

In addition to the macroeconomic policy quality measure, several control variables are incorporated to account for potential omitted variable bias. Export volume data, used to capture a country's exposure to global trade shocks, is obtained from the World Bank's

Development Indicators. Domestic credit to the private sector, expressed as a percentage of GDP, is included as a measure of financial sector development and liquidity conditions, drawn from the same source.

A composite indicator, the CPIA Average Rating, is created by calculating the average of five CPIA sub-scores: economic management, financial sector development, social inclusion and equity, property rights and rule-based governance, and social protection. This custom variable captures broader institutional quality and policy coherence beyond macroeconomic management alone.

Together, the dataset integrates key themes such as governance quality, financial market development, trade openness, and social inclusion. These themes reflect the complex, multifaceted drivers of inflation and economic stability across countries.

Table 2 – Summary Statistics

| <b><u>Observation</u></b> | <b><u>Obs</u></b> | <b><u>Min</u></b> | <b><u>Std. Dev</u></b> | <b><u>Min.</u></b> | <b><u>Max</u></b> |
|---------------------------|-------------------|-------------------|------------------------|--------------------|-------------------|
| <i>Inflation</i>          | 856               | 37.43121          | 835.0617               | -35.83668          | 24411.03          |
| <i>Macro</i>              | 922               | 3.630694          | .6785062               | 1                  | 5.55              |
| <i>Export</i>             | 866               | 242.8489          | 554.8185               | 10.24806           | 9589.121          |
| <i>CPIA_Avg</i>           | 922               | 3.152354          | .4864851               | 1.1                | 4.32              |
| <i>Credit</i>             | 847               | 32.80088          | 25.57832               | -41.18233          | 135.1             |

### Empirical Model Specification

#### Regression Model and Theoretical Justification

We employ a fixed-effects linear model with the clustered panel data framework, operationalized through an ordinary least squares (OLS) strategy. The model to be estimated is articulated as follows

$$\begin{aligned} Inflation_{it} = & \beta_0 + \beta_1 Macro_{it} + \beta_2 Export_{it} + \beta_3 Macro_{it} \times Export_{it} + \\ & \beta_4 CPI\ Avg_{it} + \beta_5 Credit_{it} + \alpha_i + \gamma_t + \epsilon_{it} \end{aligned}$$

Where  $Inflation_{it}$  represents the inflation in consumer prices for a country  $i$  at time  $t$ . The model uses  $Macro_{it}$  as its primary explanatory variable, and a proposed interaction term  $Macro_{it} \times Export_{it}$ , control variables  $Export_{it}$ ,  $Credit_{it}$ ,  $CPI\ Avg_{it}$ , and representing the other cluster control variables we use element fixed-effects  $\alpha_i$ , time fixed-effects  $\gamma_t$  and an error term  $\epsilon_{it}$

We expect the sign of the coefficient  $\beta_1$  to be negative, as effective macroeconomic management should theoretically curb the impact of inflation on consumers. Countries with higher macroeconomic policy ratings are characterized by stronger institutional frameworks and more credible policy environments, both of which are associated with lower inflation volatility.

#### Variable Selection Rationale

The primary explanatory variable of interest in this study is CPIA Macroeconomic Management Rating (Macro), an ordinal categorical measure of macroeconomic policy quality on a standardized 1–6 scale, incremented by 0.5. Across all estimated models, Macro consistently demonstrates strong statistical significance at the 1% level. For instance, in the



preferred interaction model (Column 6 of Table 3), the coefficient on Macro is -4.644 ( $p < 0.01$ ). Even in the absence of additional interaction terms or controls (Column 1), Macro remains highly significant with a coefficient of -3.222 ( $p < 0.01$ ), underscoring its centrality to the study's research question.

Export Volume (Export) is included as an independent control variable and as part of an interaction term with Macro. The rationale stems from the influence of external trade shocks on domestic inflation dynamics. While Export alone is marginally significant in the final model ( $p < 0.1$ ), the interaction term (Macro  $\times$  Export) achieves statistical significance at the 10% level (coefficient = 0.00792;  $p < 0.1$ ), meaning that countries with higher indices are expected to reduce the impact of macroeconomic policy on inflation.

CPIA Average Rating (CPIA Avg), a custom composite measure of broader institutional quality, is included to control for the potential confounding influence of a country's financial sector development, property rights protections, and social inclusion efforts. Although CPIA Average exhibits only marginal statistical significance ( $p < 0.1$ ) in the interaction model, its inclusion is critical. Omitting CPIA Average leads to substantial omitted variable bias, shifting the estimated effect of Macro significantly and reducing its interpretability. Its presence ensures that the estimated relationship between macroeconomic management and inflation is not conflated with broader governance quality effects.

Domestic Credit to Private Sector (Credit Financial), expressed as a percentage of GDP, is incorporated to account for liquidity conditions and credit market dynamics, which

existing research identifies as important determinants of inflation. This variable demonstrates consistent statistical significance at the 1% level across all specifications, further validating its inclusion.

In evaluating potential omitted variables, we systematically tested several additional controls across multiple specifications. These included GDP per capita growth, real GDP levels, depth of credit information systems, electricity access rates (as a proxy for infrastructure quality), and levels of informal payments (as a proxy for corruption). Models including GDP growth and GDP constant dollars did not yield any meaningful change in the Macro coefficient and were therefore excluded. Including depth of credit introduced slight changes but substantially reduced the number of observations, undermining estimation robustness. Similarly, including electricity access and informal payments did not produce statistically significant shifts in the key coefficients, nor did they meet the threshold of moving the estimated coefficient on Macro by more than one standard error. As such, these variables were ultimately omitted from the final models to preserve sample size and model parsimony.

Estimating quadratic and log-linear models considered the possibility of nonlinearity (Columns 3 and 4). However, none of these alternative specifications improved model fit as much as the linear interaction term, as assessed by adjusted R-squared values and statistical significance tests. Consequently, a linear model with an interaction term between Macro and Export (Column 6) was selected as the most theoretically coherent and empirically robust model for this analysis.

Table 3: - Estimates of Macroeconomic Rating: Effect on Change in Inflation across countries and time

| VARIABLES          | (1)<br>Inflation  | (2)<br>Inflation       | (3)<br>Inflation       | (4)<br>Inflation       | (5)<br>Inflation  | (6)<br>Inflation       |
|--------------------|-------------------|------------------------|------------------------|------------------------|-------------------|------------------------|
| Macro              | -269.0<br>(255.7) | -3.222***<br>(0.691)   | -10.14**<br>(4.106)    | 3.869<br>(3.375)       | -608.9<br>(564.5) | -4.644***<br>(1.104)   |
| Macro^2            |                   |                        | 0.970*<br>(0.534)      |                        |                   |                        |
| Export             |                   | 4.56e-06<br>(0.000260) | 1.29e-05<br>(0.000253) | 1.56e-05<br>(0.000253) | -6.812<br>(6.438) | -0.0317*<br>(0.0163)   |
| CPIA_Avg           |                   | 3.483**<br>(1.737)     | 3.428**<br>(1.719)     | 3.333*<br>(1.700)      |                   | 3.284*<br>(1.776)      |
| Credit Financial   |                   | -0.0773***<br>(0.0219) | -0.0688***<br>(0.0222) | -0.0691***<br>(0.0219) |                   | -0.0741***<br>(0.0215) |
| Log(Macro)         |                   |                        |                        | -24.44*<br>(12.40)     |                   |                        |
| Macro x Export     |                   |                        |                        |                        | 1.703<br>(1.610)  | 0.00792*<br>(0.00407)  |
| Constant           | 1,030<br>(943.6)  | 10.51**<br>(5.149)     | 22.36***<br>(8.049)    | 16.08***<br>(5.674)    | 2,376<br>(2,169)  | 16.67**<br>(6.451)     |
| Observations       | 856               | 794                    | 794                    | 794                    | 825               | 794                    |
| Number of country  | 79                | 75                     | 75                     | 75                     | 75                | 75                     |
| Adjusted R-squared | 0.012             | 0.039                  | 0.042                  | 0.043                  | 0.022             | 0.044                  |

Robust standard errors in parentheses

\*\*\* p&lt;0.01, \*\* p&lt;0.05, \* p&lt;0.1

### Results

The coefficient on the CPIA macroeconomic management rating in the final model is -4.644 ( $p<0.01$ ), indicating that higher macroeconomic policy quality is significantly associated with lower inflation rates. Export volume alone has a small, marginally significant negative effect on inflation ( $p=0.055$ ). Critically, the interaction term between CPIA macro rating and export volume is positive and marginally significant ( $p=0.055$ ), implying that the deflationary impact of macroeconomic policy quality weakens as export volume increases.

Domestic credit to the private sector is also shown to significantly reduce inflation ( $p < 0.01$ ), emphasizing the importance of stable financial sector conditions. CPIA average rating, although only marginally significant ( $p = 0.068$ ), proves essential; without controlling for this broader measure of social, economic, and institutional quality, the relationship between macroeconomic management and inflation loses statistical significance, indicating positive Omitted Variable Bias. Thus, a nation's broader commitment to debt management, social inclusion, and property rights is crucial for macroeconomic policy to effectively curb inflation.

Overall, better macroeconomic management is strongly associated with lower inflation. However, this effect is moderated by export volumes: countries with larger export sectors experience a weaker inflation-reducing benefit from improvements in macroeconomic policy. Financial development, as proxied by domestic credit to the private sector, also plays a significant role in containing inflation. While broader institutional quality, captured by the CPIA Average Rating, must be controlled to accurately estimate the impact of macroeconomic management, its direct relationship with inflation is positive and marginally significant. Importantly, despite its statistical significance, this model only captures around five percent of the total variance in inflation, showing that a nation's rate of inflation is not solely or primarily controlled by its economic policy, even when controlling for broader institutional differences.

### **Limitations and Future Research**

While the models explain some of the variation in inflation outcomes (Adjusted R-squared approximately 5%), a substantial portion remains unexplained. Future research

could enhance explanatory power by incorporating additional variables such as fiscal deficits, political stability indices, and more granular trade shock measures, or investigating other primary variables.

Also, while the CPIA Average Rating serves as a useful control, it introduces some multicollinearity with the macroeconomic management index. Future studies could refine these hypotheses or apply instrumental variable techniques to address potential biases.

The results of major nation outliers, namely Zimbabwe, were accounted for in this model by using country-level fixed effects; however, it is possible that extreme outliers may have biased intra-cluster results. Future studies can choose to control for this by either omitting Zimbabwe from the model or using a Mean Absolute Error model instead of Ordinary Least Squares to increase the resistance to outliers.

Lastly, the positive correlation found between the CPIA Average rating and inflation as a primary independent variable is interesting but not strictly interpretable from the results of this study. Further studies should investigate the impacts of CPIA social management and justice indices on inflation as a primary independent variable to further clarify their true relationship with inflation.

### **Conclusion**

This study finds that the quality of a nation's macroeconomic policy exerts a statistically significant influence on inflation outcomes, but that export activity levels condition this relationship. Specifically, a one-point increase in the CPIA Macroeconomic Management Rating is associated with a 4.644 percentage point decrease in inflation when

the export index is 0. However, this inflation-reducing effect weakens by 0.00407 percentage points for each additional point in the export index.

Importantly, macroeconomic policy quality only emerges as a significant predictor of inflation once broader structural factors—including domestic financial sector stability, dedication to debt management, social inclusion, and property rights protections—are controlled for. Policymakers seeking to reduce inflation cannot rely solely on macroeconomic fine-tuning; rather, effective policy must be complemented by robust financial sectors, equitable social policies, and strong institutional frameworks.

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